

LUMAFLOW · CALCULATION

LumaFlow sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-26	Amish Chadha	Budget top-up approved by Amish; rerun with budget_usd \$340, R16 now met
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); rerun for the 50 mm bore, 0.95 liner, 1.2 L/min, wall dose sensor, 15 Hz per L/min flow sensor, 10 mm window, M5 rods, pressure limiter and thermal cut-back
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (flow, Monte Carlo optics and dose, dose monitor, switching, power, thermal, pressure drop, window and structure, size, cost, options)

LumaFlow sizing calculations

This note checks the LumaFlow design as revised by LMF-DDR-002: a 50 mm bore lined with high-reflectance PTFE, a design flow of 1.2 L/min (0.32 gpm), a UV-C photodiode in the tube wall at mid height, a flow sensor of 15 Hz per L/min, an upstream pressure limiter and a thermal cut-back. On paper, LumaFlow now meets fourteen of its seventeen requirements, has one at risk (R4) and misses one (R3); one cannot be verified at TRL 3. The reduction equivalent dose (RED) in clear water (90 %/cm UV transmittance, UVT) is 51.9 (laminar) to 76.3 (plug) mJ/cm², so **R2 is now met** with the 20 % margin this note requires; in v0.1, with a 25 mm bore at 2.0 L/min, it was 14.7 to 19.0 mJ/cm². The dose at the 70 %/cm test condition is 17.4 to 21.8 mJ/cm² (**R3 not met**, kept visible by decision D1), and the parts cost \$338.00 against the \$340 budget approved by Amish on 2026-09-26 (**R16 met**). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [B5], is the line of that script's output that carries it.

Safety: These calculations concern a device meant to disinfect drinking water with UV-C light under mains pressure. They are first-principles estimates for a paper proof of concept, not a substitute for biodosimetry, datasheets or a review by a qualified engineer. A calculated dose above target is not a demonstrated dose. LumaFlow is a research and educational prototype, not a certified water treatment device. See LMF-PRC-001, Safety.

■ Scope and method

The note checks every requirement in LMF-REQ-001 v0.4 against the design in LMF-PRC-001 v0.4 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `levels()` and part volumes, so the dimensions used here are the ones in the STEP files and in drawing LMF-DWG-001 Rev P2. It reads prices from `bom/bom.csv` and the budget from `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py` (about 20 s; it runs the ray trace eight times).

Status rule for the dose: the laminar-flow RED is the design value, because the laminar profile is the more pessimistic of the two limits and the real profile lies between them [A5]. **Met** needs the laminar RED to clear the target by 20 %.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Flow	1.2 L/min (0.32 gpm), set by the restrictor; water at 20 °C for hydraulics, 25 °C for the thermal case	LMF-REQ-001 v0.4, R1
LEDs	Six 275 nm LEDs on a 32 mm circle, 60 mW UV-C each at 350 mA and 6.2 V (0.36 W, 13.02 W electrical); Lambertian emission	Typical 3535-class UV-C LED ratings; bin to be confirmed
Window	UV-grade fused silica, 57 x 10 mm on a 51 mm seat, $n = 1.496$ at 275 nm, bulk transmittance 0.995; air gap 0.8 mm; unpolarized Fresnel losses at both faces; light outside the seat aperture or the 50 mm bore is lost	Standard optics
Water	$n = 1.372$; base-e absorption $\alpha = -\ln(\text{UVT})$ per cm (0.105/cm at 90 %/cm, 0.357/cm at 70 %/cm); no scattering	UVT definition
Walls	Diffuse (Lambertian) reflection. Wetted high-reflectance PTFE liner 0.95 (sensitivity 0.80 and 0.90); wetted 316 stainless in the lower cap bore 0.30; window and LED side seen from the water 0.10	Engineering judgment for a sintered or expanded high-reflectance PTFE; to be measured
Ray trace	600,000 rays per case, 20 equal-area annuli by 60 axial bins; fluence rate = absorbed power / ($\alpha \times$ bin volume); the wall sensor window is an 8 x 8 mm absorbing patch at mid height	Collision estimator
Flow profile	Two limits: plug flow and fully developed laminar (parabolic) flow	Entrance length [A5]
Dose response	First-order inactivation of a challenge organism with 20 mJ/cm ² per log (MS2-like), with the same response at 275 nm as at 254 nm	Assumption; the wavelength response must be checked
Photodiode	SiC, 0.13 A/W at 275 nm, 0.06 mm ² active area, behind an 8 mm quartz window of 0.92 transmittance in the tube wall	Typical TO-46 SiC photodiode
Flow sensor	Hall-effect turbine, 15 Hz per L/min; Kv 0.35 m ³ /h	Pulse rate set by LMF-DDR-002; Kv to confirm
Valve	24 V DC normally closed, 4.8 W (0.2 A) held open, Kv 0.25 m ³ /h	Typical small direct-acting solenoid
Electronics	Driver 90 %; controller 0.6 W active; adapter 88 % at load, 70 % at light load, 0.1 W no-load	US DOE Level VI no-load class for small adapters
Use	15 L/day in 20 draws, 5 s run-on per draw	LMF-REQ-001
Thermal	Sink natural convection 5 W/(m ² ·K) (fins hanging down) plus radiation, $\epsilon = 0.85$; 0.5 mm pads of 3 W/(m·K); 316 at 16 W/(m·K); water film 300 W/(m ² ·K) in the lower cap (range 150 to 600); 10 K/W junction to board per LED	Engineering judgment; film coefficient is the main uncertainty
Structure	Fused quartz design tensile stress 6.8 MPa (1,000 psi) (Momentive, mechanical properties); Poisson ratio 0.17; simply supported edge; 8 bar working pressure; pressure limiter set at 4 bar	Roark circular plate, uniform load

■ A. Flow and hydraulics (R1)

The irradiated channel is 50 mm across and 242 mm long, from the window to the PTFE top disc; the flow path from inlet to outlet is 226 mm [A1]. The channel holds 475 mL, 444 mL of it between the ports [A2]. At 1.2 L/min the mean velocity is 1.02 cm/s and the mean residence time 22.2 s [A3]. The Reynolds number is 507 at 20 °C and 334

at 5 °C, so the flow is laminar [A4]. The laminar entrance length, about 1.3 m, is far longer than the flow path [A5], so the velocity profile lies between plug flow and a parabola; both limits are carried through section B. R1 is met by the 1.2 L/min restrictor, and the flow sensor range of 0.3 to 6 L/min covers it.

■ B. Optics and UV-C dose (R2, R3)

The six LEDs give 0.36 W of UV-C from 13.02 W electrical, a UV-C efficiency of 2.8 % [B1]. Of that, 87.0 % enters the water [B2]. In the 50 mm bore light crosses twice as much water between wall reflections as in the old 25 mm bore, and the liner returns 95 % of what reaches it. At 90 %/cm the water absorbs 54.0 % of the LED output and the walls 27.0 % (v0.1: 26.1 % and 49.8 %); at 70 %/cm the water absorbs 68.1 % [B3, B4].

Table 2. Dose at 1.2 L/min (mJ/cm²). Average is the flow-weighted average dose; RED is for the 20 mJ/cm² per log organism.

UVT (%/cm)	Average dose	RED, plug flow	RED, laminar flow	Reactor efficiency, laminar
95	134.3	133.7	84.5	0.63
90	76.8	76.3	51.9	0.68
85	51.7	51.3	36.8	0.71
80	37.5	37.2	27.7	0.74
75	28.4	28.1	21.7	0.77
70	22.0	21.8	17.4	0.79

At the design water quality of 90 %/cm the RED is 51.9 (laminar) to 76.3 (plug) mJ/cm² [B5]. **R2 (40 mJ/cm²) is met**, with the laminar value 30 % above the target. At 70 %/cm it is 17.4 to 21.8 mJ/cm² [B6], so **R3 is not met**, as decision D1 accepted; it is now above the Class B level of 16 mJ/cm².

The reactor efficiency (RED over average dose) is 0.68, lower than the 0.77 of v0.1: the slow laminar core on a wide bore sees less light than the water beside the wall. Streamline doses at 90 %/cm range from 42 mJ/cm² on the axis to 80 at mid radius and 1,453 beside the wall [B7]. The liner reflectance is the main optical uncertainty: 46.3, 49.9 and 51.9 mJ/cm² at 0.80 (plain PTFE), 0.90 and 0.95 [B8], so even plain PTFE meets 40 mJ/cm² but not the 20 % margin. LEDs 20 % below rating (bin, aging or temperature) give 41.5 mJ/cm² [B9].

■ C. Dose monitor (R4)

The photodiode now sits behind an 8 mm quartz window in the tube wall at mid height (Z = 166 mm), facing the enclosure. It sees 0.23 mW/cm² and gives about 16.3 nA at 90 %/cm, 2.6 nA at 80 %/cm and 0.38 nA at 70 %/cm [C1], a signal across the whole range; the top-disc position of v0.1 gave 2.4 nA at 90 %/cm and nothing resolvable at 70 %/cm. Between 90 and 85 %/cm the wall signal falls 2.9 times as fast, in log terms, as the RED [C2] (4.4 times for the old position).

Table 3. Wall sensor signal against UVT.

UVT (%/cm)	Irradiance (mW/cm ²)	Photocurrent (nA)	Signal relative to 90 %	RED relative to 90 %
95	0.738	53.0	3.25	1.63
90	0.227	16.3	1.00	1.00
85	0.084	6.1	0.37	0.71
80	0.036	2.6	0.16	0.54
75	0.012	0.9	0.055	0.42
70	0.005	0.38	0.024	0.34

The controller keeps the proportional mapping $RED_{est} = RED_{ref} \times (S/S_{ref}) \times (Q_{ref}/Q)$, which blames every loss of signal on the LEDs and so always under-reads. With $RED_{ref} = 51.9 \text{ mJ/cm}^2$ the alarm threshold is $S/S_{ref} = 0.77$ [C3]. With LEDs at rated output the alarm trips below about 88.7 %/cm UVT, and LEDs aged to 77 % of output trip it in 90 %/cm water, where the true RED is 40.0 mJ/cm^2 [C4]. A mapping that blamed the water instead would be unsafe: with LEDs at 70 % output in 90 %/cm water it would claim 46.4 mJ/cm^2 when the true RED is 36.3 [C5]. R4 stays **at risk** because the sub-nanoampere signal in poor water, the mapping and the 1 s fail-safe still need firmware and a test, which are TRL 4 work.

■ D. Flow switching and run-on (R5)

At the 0.3 L/min switch-on threshold the 15 Hz per L/min sensor gives 4.5 Hz, one pulse every 0.22 s [D1]. With the LED rise time the worst-case switch-on is 0.24 s against 0.5 s [D2] (v0.1: 0.46 s with a 7.5 Hz per L/min sensor): **R5 is met**.

During the 5 s run-on the water held in the channel receives a volume-average 19 mJ/cm^2 at 90 %/cm [D4]. Taking the dose each parcel collected while flowing plus the run-on, every parcel in the flow path has at least 62 mJ/cm^2 when flow stops (plug flow) [D3].

■ E. Power and energy (R9)

The LEDs draw 13.02 W; with the driver, the 4.8 W valve and the controller the 24 V bus carries 19.87 W, and the mains draw is 22.6 W while water flows [E1]. Standby is 0.21 W, dominated by the adapter's no-load draw and the powered flow sensor [E2]. **R9 is met** (25 W and 0.5 W).

At 15 L/day and 1.2 L/min the LEDs run 14.2 min/day (86 h/year); the unit uses 10.4 Wh/day or 3.8 kWh/year [E3]. At that rate 10,000 h of LED on-time would take 116 years [E4], so LED life is set by aging in storage and by switching, not by on-time. For comparison, tap-scale mercury units drawing 13 to 22 W all day (VIQUA VT1, VT4 and S2Q-PA spec sheet) use 114 to 193 kWh/year [E5].

■ F. Thermal (R10)

The LEDs turn 12.66 W into heat. The larger 90 x 90 mm finned sink rejects it to cabinet air through 2.46 K/W; the sink, spreader ring, lower cap and board store 802 J/K [F1]. The water path through the spreader ring and the 316 stainless lower cap is 1.55 K/W and carries 9.0 W of the 12.7 W into the water, which warms by 0.11 K [F2].

With continuous flow the LED board settles at 44.8, 40.3 or 37.1 °C for water-film coefficients of 150, 300 or 600 W/(m²·K) [F3], below 50 °C in every case; the LED junctions sit near 61 °C [F4]. The board temperature sensor and the cut-back of LED current above 50 °C (decision in LMF-DDR-002) are a backstop for the uncertain film coefficient: **R10 is met**. An acetal lower cap, with no water path, would settle at 63 °C and pass 50 °C after 31.4 min of flow [F5], which supports the 316 lower cap (LMF-DDR-002 confirms it). If the LEDs stuck on with no flow, the board would head for 53 °C, where the cut-back would act; the 5 s run-on adds only 0.08 K [F6].

■ G. Pressure drop, window and structure (R7, R8, R17)

Table 4. Pressure drop at 1.2 L/min, excluding the flow restrictor [G1, G2].

Element	Drop (bar)
Flow sensor (Kv 0.35)	0.04
Solenoid valve (Kv 0.25)	0.08
1 m of 1/4 in bore tube	0.012
Fittings (K = 6 in all)	0.012
Reactor ports	0.002
Reactor channel	below 0.00001
Total	0.15 (2.2 psi)

R8 is now worded to exclude the flow restrictor, which absorbs excess supply pressure by design. The unit drops 0.15 bar against 0.5 bar [G2]; with a Kv 0.5 sensor and a Kv 0.4 valve it would be 0.08 bar, and at the minimum supply of 2 bar, 1.85 bar remains for the restrictor and faucet [G3]. The margin covers the assumed flow coefficients, so **R8 is met**; the coefficients are still to be confirmed from the datasheets of the parts chosen.

The 57 x 10 mm window on its 51 mm seat carries a peak stress of 6.18 MPa at 8 bar, inside the 6.8 MPa design stress [G4]: **R7 is met**, with a 9 % margin (v0.1: 4.46 MPa on a 26 mm seat). Unprotected, a 16 bar water-hammer spike would raise it to 12.4 MPa. With the pressure limiter set at 4 bar the window carries 3.09 MPa static, and a transient doubling to 8 bar gives 6.18 MPa [G5]: **R17 is met** as an installation requirement. The 8 mm sensor window sees 1.69 MPa and the tube hoop stress is 8.3 MPa [G6]. Each cap carries an end load of 2,655 N; four M5 316 rods take 664 N each, 47 MPa in the thread [G7].

■ H. Size and I. Cost (R14, R16)

The unit is 277 x 100 x 322 mm without the adapter, inside the 350 x 150 x 350 mm envelope [H1]: **R14 is met**.

The 16-line BOM totals \$338.00 against the `budget_usd` of \$340 (raised from \$225 by Amish on 2026-09-26), \$2.00 under [I1]: **R16 is met**, with little margin. The largest lines are the LEDs (\$54), the high-reflectance PTFE liner (\$45) and the machined 316 lower cap (\$42) [I2]. The rise from \$249.00 comes from the larger reactor: the liner, window, caps, tube and sink all grow with the 50 mm bore.

■ J. Flow limits and options

Table 5. Flow limits for this reactor and the options against R3.

Case	Result
Six LEDs, 90 %/cm	40 mJ/cm ² (laminar) up to 1.60 L/min; 48 mJ/cm ² (Met with margin) up to 1.30 L/min [J1]
Same reactor at the former 2.0 L/min	32.7 (laminar) to 45.9 (plug) mJ/cm ² [J2]
Option B (R3 at 70 %/cm)	14 LEDs, 30 W of LED power [J3]; R9 would fail
Option C (R3 at 70 %/cm)	Six LEDs meet 40 mJ/cm ² below 0.48 L/min [J4]; proportional valve, later upgrade under D1

The 1.2 L/min design flow sits inside the 1.30 L/min limit for the 20 % margin. Filling a 1 L bottle takes about 50 s.

■ K. Requirement status

Table 6. Requirement status at TRL 3 (LMF-REQ-001 v0.4). Not met items first.

ID	Target	Value (tag)	Status
R3	RED 40 mJ/cm ² or more at 1.2 L/min, 70 %/cm	17.4 to 21.8 mJ/cm ² [B6]	Not met (accepted by D1)
R16	Parts cost \$340 or less; no custom PCB	\$338.00 [I1]; module-based electronics	Met
R4	Dose monitor; alarm and valve closed within 1 s	Wall signal 16.3 nA at 90 %/cm, 0.38 nA at 70 %/cm; alarm below about 89 %/cm [C1, C4]	At risk
R15	Window and LED head replaced in 15 min	Four tie rods and two push-fit ports; needs a build to time	Not verifiable at TRL 3
R1	1.2 L/min, restrictor	Restrictor; sensor range covers it [A3]	Met
R2	RED 40 mJ/cm ² or more at 1.2 L/min, 90 %/cm	51.9 (laminar) to 76.3 (plug) mJ/cm ² [B5]	Met
R5	LEDs on within 0.5 s above 0.3 L/min; 5 s run-on	0.24 s worst case [D2]	Met
R6	No mercury; full output in 0.1 s	LEDs, microsecond rise	Met
R7	Window 6.8 MPa or less at 8 bar	6.18 MPa [G4]	Met
R8	0.5 bar or less at 1.2 L/min, excluding the restrictor	0.15 bar, on assumed Kv [G2]	Met
R9	25 W flowing, 0.5 W standby	22.6 W, 0.21 W [E1, E2]	Met
R10	LED board 50 °C or less; cut-back above 50 °C	37.1 to 44.8 °C steady [F3]	Met
R11	Food-contact wetted parts; no UV-C on plastics but PTFE	316 lower cap; PTFE liner, top disc and 316 insert shield the acetal upper cap	Met
R12	No UV-C outside the unit; interlocks	Metal and PTFE light path; sensor window sealed by the photodiode saddle; interlock by design	Met
R13	24 V DC only at the unit	Certified adapter and input fuse	Met
R14	350 x 150 x 350 mm or less	277 x 100 x 322 mm [H1]	Met
R17	Pressure limiter at 4 bar or less upstream	Window 6.18 MPa at twice the setting [G5]	Met

Counts: 14 met, 1 not met, 1 at risk, 1 not verifiable at TRL 3 [K1].

■ Numbers changed from v0.1

Table 7. v0.1 (25 mm bore, 2.0 L/min) against v0.2 (LMF-DDR-002 design).

Quantity	v0.1	v0.2
Bore, channel, window	25 mm, 246 mm, 32 x 6 mm	50 mm, 242 mm, 57 x 10 mm [A1, G4]
Design flow	2.0 L/min	1.2 L/min [A3]
Liner reflectance (assumed)	0.80	0.95
UV-C absorbed by water at 90 %/cm	26.1 %	54.0 % [B3]
RED at 90 %/cm	14.7 to 19.0 mJ/cm ² (R2 not met)	51.9 to 76.3 mJ/cm ² (R2 met) [B5]
RED at 70 %/cm	6.3 to 7.4 mJ/cm ²	17.4 to 21.8 mJ/cm ² [B6]
Dose sensor signal at 90 %/cm and 70 %/cm	2.4 nA, unresolved	16.3 nA, 0.38 nA [C1]
Worst-case switch-on	0.46 s	0.24 s [D2]
Pressure drop	0.42 bar	0.15 bar [G2]
LED board, nominal	48.9 °C	40.3 °C [F3]
Window stress at 8 bar	4.46 MPa	6.18 MPa [G4]
Energy	3.1 kWh/year	3.8 kWh/year [E3]
Envelope	235 x 72 x 322 mm	277 x 100 x 322 mm [H1]
Parts cost	\$249.00	\$338.00 [I1]
Requirements	8 met, 4 at risk, 3 not met, 1 not verifiable (16)	13 met, 1 at risk, 2 not met, 1 not verifiable (17); 14 met and 1 not met in v0.3 after the budget top-up [K1]